

to accelerate or decelerate trading rates in real-time. Portfolio managers use volatility to evaluate overall portfolio risk, as input into optimizers, for value-at-risk calculations, as part of the stock selection process, and to develop hedging strategies. Derivatives desks use volatility to price options and other structured products. In addition, plan sponsors use volatility to understand the potential that they will or will not be able to meet their long-term liabilities and financial obligations. Volatility is a very important financial statistic.

DEFINITIONS

Price Returns/Price Change

Price returns or price change can be computed using either the “percentage” returns or “log” returns formulation. But since returns have been found to be log-normally distributed it is appropriate to use the log returns calculation. We describe these two measures:

Percentage Price Return

$$r_t = \frac{p_t}{p_{t-1}} - 1 \quad (6.1)$$

Log Price Return

$$r_t = \ln\left(\frac{p_t}{p_{t-1}}\right) \quad (6.2)$$

Where $\ln(\cdot)$ represents the natural log function.

Average Return

This average period price return $\bar{r}$ is calculated differently for the “percentage” and “log” methods.

Average return—percentage method

The process for computing an n-period average return using the percentage change methodology is:

Start with:

$$p_n = p_0 \cdot (1 + \bar{r})^n$$

Then solving for $\bar{r}$ we have:

$$\bar{r} = \left(\frac{p_n}{p_0}\right)^{\frac{1}{n}} - 1 \quad (6.3)$$